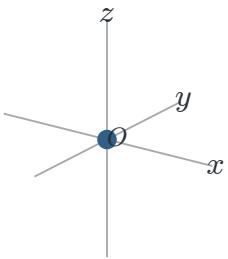
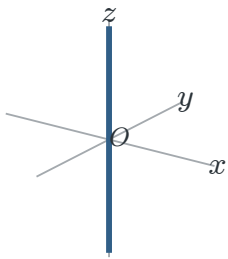
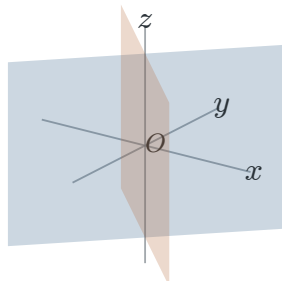
$x^2/a^2 + y^2/b^2 + z^2/c^2 = 0$



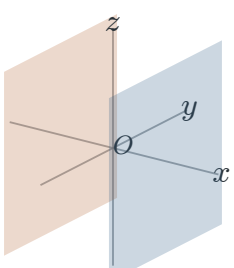
$x^2/a^2 + y^2/b^2 = 0$



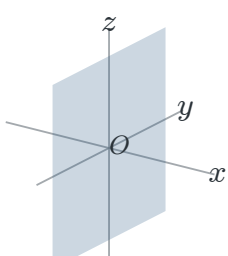
$x^2/a^2 - y^2/b^2 = 0$



$x^2 = a^2$



$x^2 = 0$



$S = \emptyset$

$x^2/a^2 + y^2/b^2 + z^2/c^2 = -1$

$x^2/a^2 + y^2/b^2 = -1$

$x^2 = -a^2$